Power systems, operations and maintenance database of Malawi's healthcare facilities

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Background

Access to reliable energy supply is crucial for delivering health services and achieving global health care access (WHO, 2015). This relates to several Sustainable Development Goals (UN, 2015). Primarily it is concerned with SDG 3 (Good Health and Well-being), since it enables lighting, sterilization, diagnostics, refrigeration and the use of medical devices, without which basic healthcare could not be delivered (Shastry & Rai, 2021; WHO & World Bank, 2023). Indirectly, it also contributes to SDG 7 (Affordable and Clean Energy) by promoting the distribution of sustainable forms of energy access and supply, especially in remote and rural areas where a grid connection would be hard to achieve. Furthermore, it influences SDG 9 (Industry, Innovation and Infrastructure) by strengthening health system infrastructure.

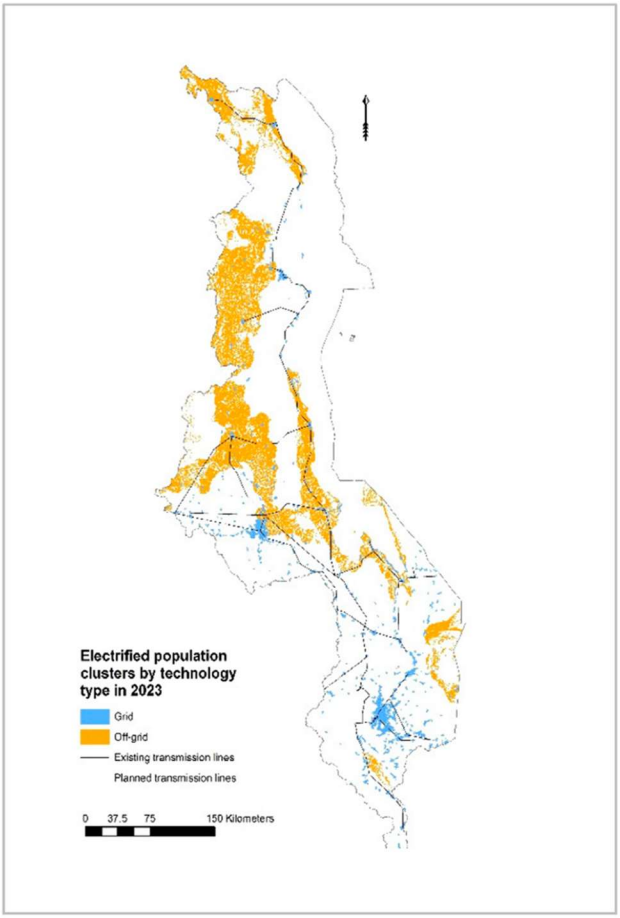

*Figure SEQ Figure * ARABIC 1 Source: https://www.mdpi.com/energies/energies-12-01395/article_deploy/html/images/energies-12-01395-g008.png*

Even though the importance of energizing HCF seems natural, data on the topic remains scarce and fragmented (SETI, 2018). This consequently limits the ability of governments, development partners, and service planners to make informed decisions, track progress, or prioritize interventions effectively. The majority of studies to address energy access have focused primarily on household settings, while comprehensive data on healthcare facilities remain limited (Jeuland et al., 2021; SETI, 2018). Work on energizing HCF in Africa is especially rare. Indeed, several African nations do not appear to have a single quantitative peer-reviewed study that investigates an energy-related question (Jeuland et al., 2021). In sub-Saharan Africa, one in four HCF is entirely without electricity, most of them are located in rural areas. Of the ones that have electricity, three-quarters do not have reliable power supply. (Adair-Rohani et al., 2013; Moner-Girona et al., 2021).

Malawi, a landlocked country in southeastern Africa, relies heavily on hydropower from Lake Malawi, which generates over 95% of the nation's electricity (Ngongoma et al., 2025). Despite this,

it has one of the lowest electricity access rates globally, with just 14% of the population having access in general and only 11% connected to the grid (Ehimen et al., 2023). Considering only rural areas, this percentage decreases to only 4% while 42% of urban residents have access to electricity. With a rapidly growing population of over 19 million, expanding at an annual rate of 2.8%, the rate of electrification continues to fall behind population growth (World Bank, 2019).

Malawi currently has 1920 healthcare facilities (HCF), both public and private according to the Malawi Ministry of Health (*MHFR*, 2025). In Malawian HCF electricity is often available but with frequent blackouts disrupting service delivery. Similar to the rest of sub-Saharan Africa, data on this issue is scarce, as only a handful of studies have been conducted. One study from 2020 found that

*Figure SEQ Figure * ARABIC2 PV system in a Malawian healthcare facility. Source: https://www.victronenergy.com/blog/2022/07/29/changing-health-outcomes-in-malawi/*

82% of Malawian HCF had grid access, 26% had other electricity sources such as solar and only 2% had no access to any electricity at all. The latter were mostly primary-level facilities, particularly rural health posts (Reuland et al., 2020). This was also found by another study that analyzed data from the 2018/2019 Harmonized Health Facility Assessment (HHFA)(WHO, 2023). They found that rural HCF had the most unstable electricity access, partly because they rely on off-grid energy sources four times more often (Ngongoma et al., 2025). Most commonly rural facilities are primarily-level facilities so it is unsurprising that the same trend was observed comparing primary-level facilities to secondary and tertiary-level facilities

Because the grid expansion is not progressing quick enough, off-grid systems will be the realistic focus in rural areas. (Franco et al., 2017) give a very good introduction and review of off-grid energy supply and storage systems, including hybrid systems, in the context of the global south. Generally hybrid solar systems seem to fit the bill best in most cases. During periods of low demand, renewable energy systems are used to charge storage systems, primarily batteries. In the event of an outage, these renewable energy systems, along with the batteries, provide the required electricity. During

extended outages where the stored energy and renewable electricity are insufficient, generators can be utilized as a final resort, minimizing their associated high operating cost. When expanded with larger energy supply and sufficient battery storage solutions, such systems can even power multiple community entities in the vicinity (N. Kassam, 2024).

The continuous operation of standalone energy systems in HCF presents a persistent challenge. Poor maintenance leads to frequent breakdowns and repairs are often delayed, primarily due to insufficient maintenance staff, supplies, transport and funding (Franco et al., 2017; Shadvar & Rahman, 2024). In 2011, seventy four percent of all health financing in Malawi comes from donors (Borghi et al., 2018), so when energy systems responsibilities are handed over to national partners, they often lack the necessary human and financial resources to properly manage operations and maintenance (O&M). This system frequently leads to the premature failure of the equipment (Sharma et al., 2024). The volatility of donor funding thus contributes to a weakness in organizing support in O&M of energy systems as well as medical devices (Borghi et al., 2018; World Bank, 2019). Coordination problems also seem to arise more often when donors co-finance projects (Marineau & Findley, 2020). In many cases, the handover process and the definition of roles and responsibilities among stakeholders are not clearly established during project planning and implementation. As a result, operation and maintenance activities are neglected, since no specific agency is assigned the responsibility (Sharma et al., 2024).

Maintenance and capacity building for effectively implementing off-grid solar systems in rural health facilities remain a priority, and related data on financiers and O&M strategies for Malawi is missing. In an effort to get an overview of donations, Malawi's "Open Aid Map" was created, listing countrywide health projects (Weaver et al., 2013). This data however, is not updated and only indicates whether a project had received assistance to this point (2013). To understand the correlation between donors and long-term effectiveness of their respective donations requires data on the current state of functionality and maintenance of these systems.

Problem statement and goals

Few studies exist that gather detailed data on power systems in HCF in rural Malawi. This hinders governments, development partners, energy engineers and service planners from making data-informed decisions, tracking progress and prioritizing interventions.

A large percentage of power systems and medical devices come from donations. Sometimes the approach to donations seems to be that the upfront financing and short-term O&M are covered, while the facilities are expected to cover long-term costs and responsibility without sufficient institutional or governmental support.

These problems are most prevalent in rural areas, in facilities that provide primary care level services. This includes health posts, dispensaries, maternity clinics, health centers and community hospitals. Primary level facilities account for over 90% of Malawi's health system (World Bank Group, 2019).

Goals

- Collect detailed, facility-level data on power supply status, including energy source mix, nameplate characteristics and outage protection.
- Document the origins and current O&M structure of these facilities' energy systems and key medical devices. Then draw up a policy white paper that provides an overview of successes and failures and offers a strategy for improving accountability among stakeholders moving forward

Energy systems, operations and maintenance dataset of Malawi's health posts

Facility-level data on the energy supply status will be collected. This means documenting the energy source and storage mix by 1) coordinating with facility administrators to understand the system's energy topology, 2) documenting nameplate characteristics of all components and the source(s) of their investment, and 3) inspecting functionality. Additionally, the origins and current operations and maintenance of energy systems and key medical devices are documented by talking to responsible persons. The goal is then for this dataset to enable accountability among associated donors or implementing partners for implemented energy systems.

Research Questions

The research questions addressed discussed challenges are:

1. What proportion of primary care facilities in Malawi have functioning power systems, and what types of primary and backup energy sources, energy storage, and surge protection are currently in place?
2. Among energy systems and medical devices, what percentage remain functional, and which stakeholders are accountable for their deployment, ownership, and ongoing maintenance?
3. How do the characteristics of power systems (e.g., energy source mix, functionality, maintenance) differ by region, catchment population size, and associated donors or implementing partners?
4. Does the health post have an established plan to reduce their loads if not all demand can be met. (From original n-line project proposal)

Optional

5. What is the relationship between power systems and the functionality of critical medical equipment (e.g., ultrasound machines but probably needs to be something cheaper)?
    - *Inspired by: What is the association between power supply and the availability and functionality of eg. ultrasound (USS) equipment in Malawi's healthcare facilities.*(Ngongoma et al., 2025)

Methods

Implementation

Data will be collected between July and October 2025, the dry season in Malawi. Along with Till Häussner, a fellow Global Health Engineering thesis student, I will travel to 2-3 HCF per week by car and expect to spend one day at each facility. I will focus on studying facilities that offer primary care. Which specific facilities we will visit is still to be determined and will depend on the availability of the local officials and responsible persons.

Data Collection

1. **Power system**

Qualitative and Quantitative data will be collected on various characteristics of power supply status, including the energy source/storage mix, nameplate characteristics and outage preparation and protection. Categories are based on the ones provided by (Ngongoma et al., 2025; Reuland et al., 2020; Suhlrie et al., 2018). Accessing this information will require talking to a responsible person; however, I can collect nameplate characteristics and other related energy metrics without additional support once pointed in the right direction.

2. **Accountability**

Qualitative data will be collected on O&M and original financing mechanism of the power supply and key medical devices via a questionnaire. This will require a meeting with one or two knowledgeable responsible persons, preferably on-site.

The questionnaire will primarily be based on the power related survey questions from WHO's harmonized health facility assessment (HHFA)(WHO, 2023). They can be found in appendix A.

The questionnaire will also be inspired by the questions for administrators described in the (Reuland et al., 2020) "Supplementary Materials". It is semi-structured and questions are organized into sections. A preliminary adapted version of the Interview can be found in appendix B.

The questionnaire is still work in progress and is set to be improved in the coming weeks.

Planning

Gantt chart

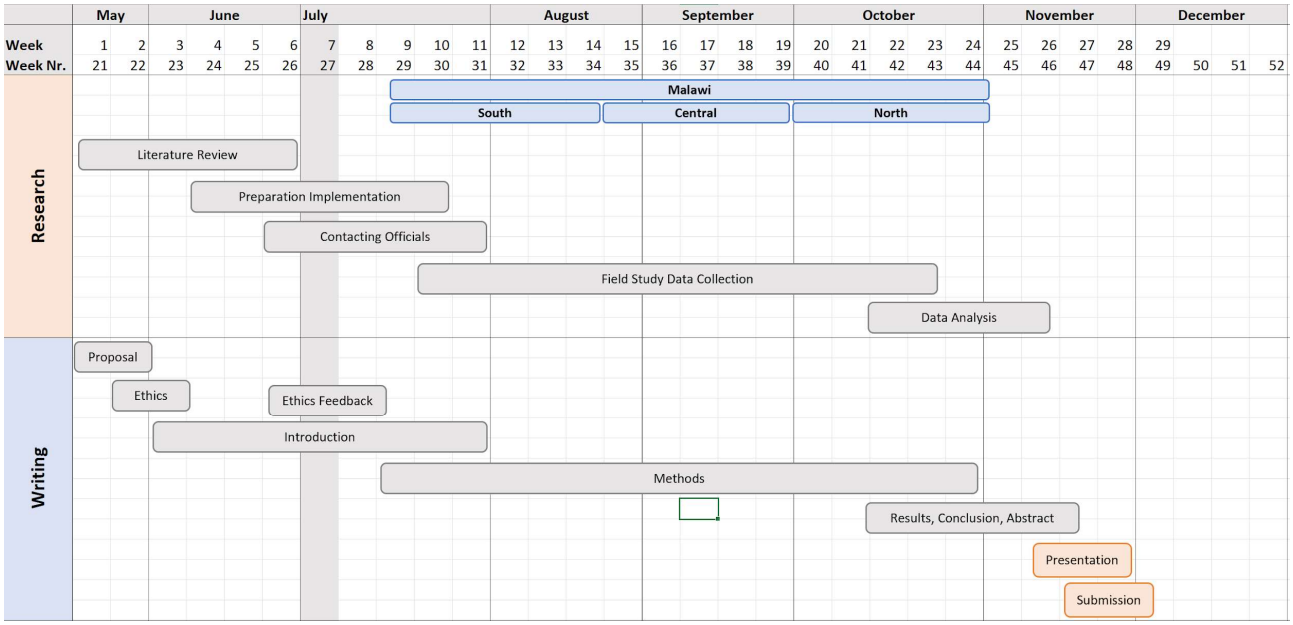

Figure 3. Gantt chart for research work and thesis writing.

Data Management Plan (DMP)

In this section the DMP is provisionally laid out but is subject to change over the course of the thesis.

Data collection

dataset	**data type**	**data format**	**data documentation**
Health post metadata	*Secondary data*	*Source → Google Sheet → CSV export → GHE GitHub*	*Information on HCF in Malawi like name, location, type etc.*
Health post metadata	*Observational data*	*Paper-based → Google Sheet → CSV export → GHE GitHub*	*Information on HCF in Malawi like name, location, type etc.*
Power system status	*Observational data*	*Google Sheet → CSV export of raw data → GHE GitHub*	*preliminary variables → see table below. The dataset will have observations of approx. 30 HCF*
Outage protection, critical loads, load reduction	*Qualitative semi-structured questionnaire responses*	*Google Forms → Google Sheets → CSV export of raw data →*	*See Interview in Appendix*

		GHE GitHub	
Origins & O&M	*Qualitative semi-structured questionnaire responses*	*Google Forms → Google Sheets → CSV export of raw data → GHE GitHub*	*See Interview in Appendix*

Health Post Information

Variable	**Options**
Name	-
Region	North, center, south
Facility Type	Health post, dispensary, maternity clinic, health centers, community hospital
Facility Ownership	Government, CHAM, Other

Power System Status

This proposed data structure is still in progress and set to be improved in the coming weeks.

Variable	**Options**
Energy Supply Type	MC: Grid, mini-grid, PV, wind, generator, other (wood, coal, gas)
Battery Energy Storage	Yes, no
For all selected Types in prev. 2 rows	Functional, nonfunctional, primary, backup
Implementation Date	DD/MM/YYYY
Rated Power Output	Max. output in Watts
Electrical Protection	MC: SPD, grounding, fuses, RCD, voltage relay
Batteries	
Rated Storage Capacity	kWh
Nominal Voltage	V
PV	
Surface Area	m^2
Support structure quality (UNICEF, 2018)	MC: Aeration, robust, maintenance friendly

Data storage

All generated data listed above is stored in Google Workspace within the provided folder structure. Backups of the raw data are made regularly (after each new entry) to the personal Microsoft OneDrive in a protected MS Excel sheet to guarantee its unedited state.

Budget

No	expense	amount	unit price [CHF]	total price [CHF]
1	Rental car	1	5 per day	450 for 3 months
2	Fuel	15 (weeks)	20 per week	300
3	Accommodation (months)	3.5	500	1750
4	Flights	1	1800	1800
5	Visas & Insurance	1	200	200
6	Health – vaccines, medication ...	-	-	500
7	Refreshments for Interview	40	5	200
	Total			**4700**

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Appendix A

		12.1.2. POWER SUPPLY		
R_C	1206	Does this facility have electricity from any source such as electricity grid, generator, solar or other source, including for stand-alone devices such as those used to maintain the EPI cold chain?	YES..... 1 NO 2	➔Q1210
R_C	1207	What is the facility's main source of electricity? IF RESPONSE IS DIFFERENT FOR IN- AND OUTPATIENT SERVICE SITES, PROVIDE RESPONSE FOR INPATIENT SERVICE SITES. [COUNTRY ADAPT]	CENTRAL SUPPLY OF ELECTRICITY (E.G. NATIONAL OR COMMUNITY GRID) ..... 1 GENERATOR (FUEL OR BATTERY-OPERATED) ..... 2 SOLAR-POWERED SYSTEM..... 3 OTHER ..... 6 _____ (SPECIFY)	
R_C	1208	Other than the main source, does the facility have a backup source of electricity?	YES..... 1 NO 2	
R_C	1209	During the past 7 days, was electricity available, from the main or any backup source, at all times the facility was open for services?	ALWAYS AVAILABLE (NO INTERRUPTIONS)..... 1 OFTEN AVAILABLE (SOME INTERRUPTIONS OF LESS THAN 2 HOURS PER DAY) ..... 2 SOMETIMES AVAILABLE (FREQUENT OR PROLONGED INTERRUPTIONS OF MORE THAN 2 HOURS PER DAY) 3	
R_C	2513	Is the surgical unit supported by a back-up power supply if there is a gap in the primary electricity supply?	YES.....1 NO 2	
R_C	2618	Is there electricity in this service site that is functioning now? IF YES, VERIFY ELECTRICITY IS FUNCTIONAL.	YES, OBSERVED ..... 1 NO, NOT TODAY..... 2 NO, NEVER HAVE ELECTRICITY..... 3	➔Q2620
R_C	2619	Is the emergency service site supported by a back-up power supply if there is a gap in the primary electricity supply?	YES ..... 1 NO..... 2	
R_C	2903	Is there electricity in this service site that is functioning now? IF YES, VERIFY ELECTRICITY IS FUNCTIONAL.	YES, OBSERVED ..... 1 NO, NOT TODAY ..... 2 NO, NEVER HAVE ELECTRICITY 3	➔Q2907
R_C	2904	Does the laboratory have a back-up source of electricity when the main electricity is not functioning?	YES ..... 1 NO 2	
R_C	2905	At any time during the past 7 days has the electricity for the laboratory been off for more than 2 hours at a time?	YES ..... 1 NO 2	

Appendix B

This is a heavily modified version of the Interview by Reuland et al.:

General information

1. *Asking about Health post metadata if missing*

Energy Supply

2. *What does the energy system consist of?*
    a. *What is your primary/secondary energy source?*
    b. *Is there an emergency backup like a generator?*
3. *When the primary energy source fails how do you cope?*
    a. *Do you have to prioritize certain services when using this backup supply?*
    b. *[If yes] Prompt: How do you select the services which are prioritized?*
    c. *Prompt: What do you think is necessary to maintain an adequate energy supply?*
    d. *Prompt: What do you think is the main reason for inadequate or unreliable energy supply?*

Origins, O&M, Accountability

4. *Who installed the system?*
    a. *When was the system installed?*
    b. *Have they provided maintenance services?*
    c. *[If yes] For what period of time?*
5. *Are you satisfied with the maintenance of the energy system in your facility? Why, why not?*
    a. *Who is responsible for maintaining the energy system?*
    b. *Prompt: Can you describe their duties?*
    c. *Prompt: How could maintenance be improved?*